Complex Number-1

Iota (i) and its powers :

$$i = \sqrt{-1}$$

$i^1 = i$
$i^2 = -1$
$i^3 = -i$
$i^4 = 1$

* Sum of 4 consecutive powers is always zero.
$$i + i^2 + i^3 + i^4 = 0$$
$$i^r + i^{r+1} + i^{r+2} + i^{r+3} = 0$$
* For greater powers of Iota, divide power by 4 and raise remainder as new power.

Eg: Find the value of
$$\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} + 2$$

Sol^n
$$\frac{i^{10}\left(i^{582} + i^{580} + i^{578} + i^{576} + i^{574}\right)}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} + 2$$
$$= i^{10} + 2 \Rightarrow i^2 + 2$$
$$= -1 + 2 \Rightarrow \boxed{1}$$ Ans

$\frac{4\sqrt{10}}{2}$

Definition of Complex Number :

Any number that can be represented in following form is a complex number.

$$\boxed{z = a + ib} \quad (a, b \in R)$$

* Complex Numbers are the superset of all the other classification of numbers.

$$[N \subset W \subset Z \subset Q \subset R \subset C]$$

$z = a + ib$

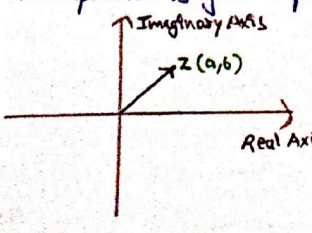
Purely Real	Purely Imaginary	Imaginary
If $b = 0$	If $a = 0$	If $b \neq 0$
Eg: $2 + i(0)$	Eg: $0 + 7i$	Eg: $1 + 2i$
$-2 + i(0)$	$0 + \pi i$	$0 + 11i$
$5.1 + i(0)$	$0 - 4.6i$	$0 - \frac{7}{11}i$

NOTE: $0 + (0)i$ is → Purely real ($a=0$) → Purely Imaginary ($b=0$) → but is not Imaginary (bcz $b=0$)

NOTE: $\sqrt{a} \cdot \sqrt{b} \neq \sqrt{ab}$, when a & b, both are negative.
Rest all cases, $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$

Geometrical presentation of CN :

* Complex no. is just a point on 'Argand Plane'



$z = a + ib$
$Re(z) = a$
$Im(z) = b$

* For complex numbers
$$\boxed{z_1^2 + z_2^2 = 0 \not\Rightarrow z_1 = z_2 = 0} \quad (\times)$$
While, in case of real no., we can simply
$$\boxed{a^2 + b^2 = 0 \Rightarrow a = b = 0} \quad (\checkmark)$$

Algebra of Complex Numbers :

Let $z_1 = 3 + 4i$ and $z_2 = 4 - i$, then

1) $z_1 + z_2 = (3+4) + i(4-1) \Rightarrow 7 + 3i$

2) $z_1 - z_2 = (3-4) + i(4+1) \Rightarrow -1 + 5i$

3) $z_1 \times z_2 = (3+4i)(4-i) = 12 - 3i + 16i - 4i^2$
$= 12 + 4 + 13i$
$= 16 + 13i$

4) $\frac{z_1}{z_2} = \frac{(3+4i)}{(4-i)} \times \frac{(4+i)}{(4+i)} = \frac{12 + 3i + 16i + 4i^2}{4^2 - (i)^2}$
$= \frac{8 + 19i}{16 + 1} \Rightarrow \left(\frac{8}{17}\right) + \left(\frac{19}{17}\right)i$

Important Values :
$(1+i)^2 = 2i$ & $(1-i)^2 = -2i$

Equality of Complex Numbers :

* Inequality b/w CN does not exist bcz 'points' can't be compared.

* If two complex numbers are equal
$$z_1 = z_2$$
$$a_1 + ib_1 = a_2 + ib_2$$
$$\Rightarrow \boxed{a_1 = a_2} \text{ and } \boxed{b_1 = b_2} \quad (\cap)$$

Ques If $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{i+1}{i-1}\right)^{n/3} = 1$, then greatest common divisor of m & n is ____.

Sol^n
$$\left(\frac{(1+i)^2}{(1-i)^2}\right)^{\frac{m}{4}} = \left(\frac{(1+i)^2}{(1-i)^2}\right)^{\frac{n}{6}} = 1 \quad (\text{Least value of})$$
$$\left(\frac{2i}{-2i}\right)^{\frac{m}{4}} = \left(\frac{2i}{-2i}\right)^{n/6} = 1$$
$$\Rightarrow (-1)^{m/4} = (-1)^{n/6} = 1$$
Since we want least possible value,
$$\frac{m}{4} = 2 \Rightarrow \boxed{m = 8} \quad ! \quad \frac{n}{6} = 2 \Rightarrow \boxed{n = 12}$$
$$\boxed{GCD(8, 12) = 4}$$ Ans

Ques The least +ve integer n such that $\frac{(2i)^n}{(1-i)^{n-2}}$ is also a +ve integer is ____.

Sol^n
$$= \frac{(2i)^n}{(-2i)^{\frac{n-2}{2}}} \Rightarrow \frac{(2i)^n}{(-1)^{\frac{n-2}{2}}(2i)^{\frac{n-2}{2}}}$$
$$= \frac{(2i)^{\frac{2n}{2}}}{(-1)^{\frac{n-2}{2}}} \Rightarrow \frac{-4}{(-1)^{\frac{n-2}{2}}}$$

$2i$ can be +ve, only when its power is 4 or its multiple.
$$\Rightarrow \frac{n+2}{2} = 4 \Rightarrow \boxed{n = 6}$$ Ans

Ques If real part of $z = \frac{3 + 2i\cos\theta}{1 - 3i\cos\theta}$, $\theta \in (0, \pi/2)$ is zero, then find $(6\sin^2 3\theta + \cos^2\theta)$.

Sol^n
$$z = \frac{(3 + 2i\cos\theta)(1 + 3i\cos\theta)}{1^2 - (3i\cos\theta)^2} = \frac{3 - 6\cos^2\theta + \ldots}{1 + 9\cos^2\theta}$$
$$Re(z) = \frac{3 - 6\cos^2\theta}{1 + 9\cos^2\theta} = 0 \Rightarrow \cos^2\theta = \frac{1}{2}$$
$$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \boxed{\theta = \frac{\pi}{4}}$$
So,
$$\cos^2\theta + \sin^2(3\theta) = \sin^2\left(\frac{3\pi}{4}\right) + \frac{1}{2}$$
$$= \boxed{1}$$ Ans

Square Root of Complex Number :

Let, $\sqrt{a + ib} = x + iy$

then,
$$\boxed{x^2 = \frac{\sqrt{a^2+b^2} + a}{2}} \qquad \boxed{y^2 = \frac{\sqrt{a^2+b^2} - b}{2}}$$

Eg: $\sqrt{3 + 4i}$ $\quad x^2 = \frac{5+3}{2} \Rightarrow 4 \Rightarrow \boxed{x = 2}$
$[\sqrt{3^2 + 4^2} = 5]$ $\quad y^2 = \frac{5-3}{2} \Rightarrow 1 \Rightarrow \boxed{y = 1}$

So, $\boxed{\sqrt{3 + 4i} = \pm(2 + i)}$ Ans

Eg: $\sqrt{-7 + 24i}$ $\quad x^2 = \frac{25 - 7}{2} = 9 \Rightarrow \boxed{x = 3}$
$[\sqrt{7^2 + 24^2} = 25]$ $\quad y^2 = \frac{25 - (-7)}{2} = 16 \Rightarrow \boxed{y = 4}$

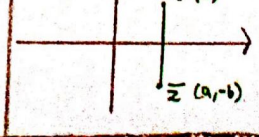
$$\boxed{\sqrt{-7 + 24i} = \pm(3 + 4i)}$$ Ans

Ques The imaginary part of $(3 + 2\sqrt{-54})^{\frac{1}{2}} - (3 - 2\sqrt{-54})^{\frac{1}{2}}$ can be :

(A) $-\sqrt{6}$ (B) $-2\sqrt{6}$ (C) 6 (D) $\sqrt{6}$

Sol^n $z = \sqrt{3 + 6\sqrt{6}i} - \sqrt{3 - 6\sqrt{6}i}$
$$z = \left[\pm(3 + \sqrt{6}i)\right] - \left[\pm(3 - \sqrt{6}i)\right]$$
$Im(z)$
0 (x) $\quad -2\sqrt{6}$ (✓) (B) Ans

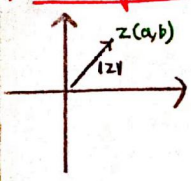
Conjugate of CN :



* It is mirror image of z in Real Axis

If $z = a + ib$ [Replace, $i \to -i$]
$\Rightarrow \bar{z} = a - ib$

Modulus of CN:

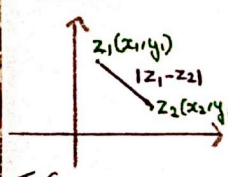


* It's distance of z from origin.

$z = a+ib$

$\Rightarrow \boxed{|z| = \sqrt{a^2+b^2}}$

* Mod of diff of two CN gives distance b/w them.



$z_1 = x_1 + iy_1$

$z_2 = x_2 + iy_2$

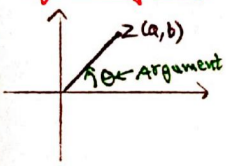
$\boxed{|z_1 - z_2| = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}}$

Thus

$|z-i| \Rightarrow$ Distance b/w 'z' & 'i'

$|z+1| \Rightarrow$ Distance b/w 'z' & '-1'

Argument of CN:



Anti clockwise $\theta \Rightarrow$ +ve

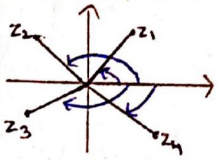
clockwise $\theta \Rightarrow -ve$

* Angle made by CN with Real axis is called its Argument.

* There can be multiple values of Argument for a CN.

$\arg(z) = \theta, \theta+2\pi, \theta+4\pi, - - - -$

Principal Argument :→



Principal Argument $\in (-\pi, \pi]$

Steps to find Principal Argument

1) For $z = a+ib$, find α as

$\boxed{\tan\alpha = \left|\frac{b}{a}\right|}$

2) Identify the quadrant of z.

3) Depending upon quadrant, its principal argument can be

I st quad. $\Rightarrow \alpha$	III rd quad. $\Rightarrow \alpha - \pi$
II nd quad. $\Rightarrow \pi - \alpha$	IV th quad. $\Rightarrow -\alpha$

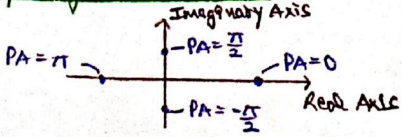
Eg: $z = 1 - \sqrt{3}i$

Sol^n 1) $\tan\alpha = \left|\frac{-\sqrt{3}}{1}\right| \Rightarrow \alpha = \frac{\pi}{3}$

2) $(1,-3) \Rightarrow$ IV th quad.

3) Principal Arg. $= -\frac{\pi}{3}$

Principal Arguments on the axes :→



Properties of Modulus, Conjugate & Argument:

(I) Properties of Modulus :→

1) $|z| \geq 0$ → If $|z| = 0$, it implies $z = 0+0i$ no other possibility

2) $|z_1||z_2| = |z_1 \cdot z_2|$

3) $|z^n| = |z|^n$

4) $|z| = |\bar{z}| = |-z| = |-\bar{z}|$

5) $\boxed{z \cdot \bar{z} = |z|^2}$ → If $|z| = 1$ i.e z → unimodular then $\boxed{z = \frac{1}{\bar{z}}}$ and $\boxed{\bar{z} = \frac{1}{z}}$

(II) Properties of Conjugate :→

1) $\overline{(\bar{z})} = z$

2) $\overline{(z_1+z_2)} = \bar{z_1} + \bar{z_2}$

3) $\overline{(z_1 \cdot z_2)} = \bar{z_1} \cdot \bar{z_2}$

4) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z_1}}{\bar{z_2}}$

5) $\boxed{z+\bar{z} = 2Re(z)}$ → If $z+\bar{z} = 0 \Rightarrow Re(z) = 0$ i.e z → Purely Imaginary and $\boxed{z = -\bar{z}}$

6) $\boxed{z-\bar{z} = i2Im(z)}$ → If $z-\bar{z} = 0 \Rightarrow Im(z) = 0$ z → Purely Real and $\boxed{z = \bar{z}}$

(III) Properties of Argument :→ (Similar to log)

1) $\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2) \pm 2k\pi$

2) $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2) \pm 2k\pi$

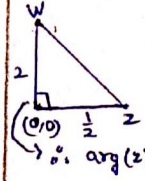
3) $\arg(z^n) = n \cdot \arg(z) \pm 2k\pi$

4) $\arg(\bar{z}) = -\arg(z) \pm 2k\pi$

Ques Let a CN, $w = 1 - \sqrt{3}i$. Let other CN z be such that $|zw| = 1$ and $\arg(z) - \arg(w) = \pi/2$. Then area of the Δ with vertices as origin, z & w is ____.

Sol^n $|w| = (1^2 + (-\sqrt{3})^2)^{1/2} = 2$

$\because |zw| = 1 \Rightarrow |z| = \frac{1}{|w|} = \frac{1}{2}$



Area of $\Delta = \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times \frac{1}{2} \times 2 \Rightarrow \frac{1}{2}$ sq unit Ans

$\because \arg(z) - \arg(w) = \pi/2$

Representation of Complex Numbers:

1) Cartesian Form :→

$\boxed{z = x+iy}$ → (x,y) → Coordinates of z on Argand plane

2) Trigonometric/Polar Form :→

$\boxed{z = r(\cos\theta + i\sin\theta)}$ Also, $\boxed{z = r\,cis\,\theta}$

r → modulus value of z

θ → Argument of z

3) Exponential/Euler Form :→

$\boxed{z = re^{i\theta}}$ r → modulus of z, θ → argument of z

Euler's Identity ⇒ $\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$

Note:

(I) $e^{-i\theta} = \cos\theta - i\sin\theta$

(II) If somewhere 'i' is with cos instead of sin, then take 1 in product with sin as $(-i^2)$

i.e $= (1)\sin\theta + i\cos\theta$

$= (-i^2)\sin\theta + i\cos\theta$

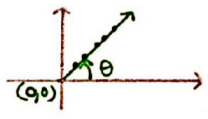
$= i(\cos\theta - i\sin\theta) = ie^{-i\theta}$

4) Vector Form :→

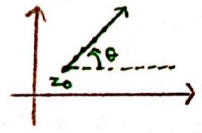
$\boxed{z = x\hat{i} + y\hat{j}}$ → (x,y) → Coordinates of z in Argand plane.

Locus of z:

* $\boxed{\arg(z) = \theta}$ → Represents a ray from origin, making θ angle with real axis.



$\boxed{\arg(z-z_0) = \theta}$ → Represents a ray from z_0, making θ angle with horizontal.

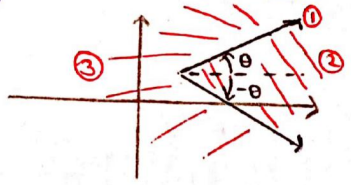


If,

$|\arg(z-z_0)| = \theta \Rightarrow \arg(z-z_0) = \pm\theta$ (Area 1) 2 Rays)

$|\arg(z-z_0)| \leq \theta \Rightarrow -\theta \leq \arg(z-z_0) \leq \theta$ (Area 2, Inside Rays)

$|\arg(z-z_0)| \geq \theta \Rightarrow$ All area outside rays. (Area 3)



* $\boxed{|z| = r}$ → Circle of radius r with centre at origin (0,0)

$\boxed{|z-z_0| = r}$ → Circle of radius r with centre at z_0.

If, $|z-z_0| = r \Rightarrow$ on circle ①

$|z-z_0| < r \Rightarrow$ inside circle ②

$|z-z_0| > r \Rightarrow$ Outside circle ③



Ques Three sets defined as

$S_1 = \{z \in C : |z-1| \leq \sqrt{2}\}$

$S_2 = \{z \in C : Re((1-i)z) \geq 1\}$

$S_3 = \{z \in C : Im(z) \leq 1\}$

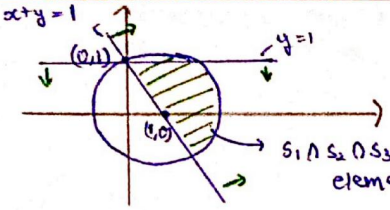
How many elements in $S_1 \cap S_2 \cap S_3$?

Sol^n ① $|z-1| \leq \sqrt{2}$ → on + inside circle of radius $\sqrt{2}$ and centre (1,0)

② $(1-i)(x+iy) = (x+y) + i(y-x) = z'$

$Re(z') = \boxed{x+y \geq 1}$ ← line passing through (0,1) & (1,0)

③ $y \leq 1 \Rightarrow$ simple line.



$S_1 \cap S_2 \cap S_3$ has ∞ elements.

De-Moivre's Theorem (DMT):

This theorem is applied when a CN is raised to some higher or fractional power.

Eg z^n, $z^{p/q}$ $\Rightarrow$ DMT

Statement-01 ⇒

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin(n\theta)$$

↳ $(e^{i\theta})^n = e^{i(n\theta)}$

Statement-02 ⇒

$$(\cos\theta + i\sin\theta)^{p/q} = \cos\left(\frac{p\theta+2k\pi}{q}\right) + i\sin\left(\frac{p\theta+2k\pi}{q}\right)$$

↳ where $k = (0, 1, 2 \dots q-1)$

Ques Find value of $\left[\frac{1+\sin\left(\frac{2\pi}{9}\right) + i\cos\left(\frac{2\pi}{9}\right)}{1+\sin\left(\frac{2\pi}{9}\right) - i\cos\left(\frac{2\pi}{9}\right)}\right]^3$

Sol^n $\frac{\pi}{2} - \frac{2\pi}{9} = \frac{9\pi-4\pi}{18} \Rightarrow \frac{5\pi}{18}$

Now,

$$z = \left(\frac{1+\cos\frac{5\pi}{18} + i\sin\frac{5\pi}{18}}{1+\cos\frac{5\pi}{18} - i\sin\frac{5\pi}{18}}\right)^3 \quad \left[\text{let } \frac{5\pi}{18} = \theta\right]$$

$$= \left[\frac{2\cos^2\frac{\theta}{2} + i\left(2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}\right)}{2\cos^2\frac{\theta}{2} - i\left(2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}\right)}\right]^3 \quad (\text{Take } 2\cos\tfrac{\theta}{2} \text{ common})$$

$\left(\because \sin\theta = 2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2} \;\&\; 1+\cos\theta = 2\cos^2\frac{\theta}{2}\right)$

$$= \left(\frac{\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}}{\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}}\right)^3 \Rightarrow \left(\frac{e^{i\frac{\theta}{2}}}{e^{-i\frac{\theta}{2}}}\right)^3 \Rightarrow e^{3i\theta}$$

$$= e^{i\left(\frac{5\pi}{6}\right)} \Rightarrow \cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)$$

$$\Rightarrow \boxed{z = -\frac{\sqrt{3}}{2} + i\left(\frac{1}{2}\right)} \quad \text{Ans}$$

Ques If $(\sqrt{3}+i)^{100} = 2^{99}(p+iq)$, then find $p \& q$.

Sol^n $\sqrt{3}+i = 2e^{i(\pi/6)}$

① $\tan\alpha = 1/\sqrt{3} \Rightarrow \alpha = \pi/6$
② $(\sqrt{3}, 1) \Rightarrow$ 1st quad.
③ $PA = \alpha = \pi/6$

So, $z = \left(2e^{i\pi/6}\right)^{100}$

$= 2^{100}\cdot e^{i\left(\frac{50\pi}{3}\right)} \Rightarrow 2^{100}\left[\cos\frac{50\pi}{3} + i\sin\frac{50\pi}{3}\right]$

$= 2^{100}\left[-\frac{1}{2} + i\left(\frac{\sqrt{3}}{2}\right)\right] \Rightarrow 2^{99}\left(-1+\sqrt{3}i\right)$

So, $\boxed{p=-1}$ & $\boxed{q=\sqrt{3}}$ Ans

Cube Root of Unity:

We can write,

$1 = 1 + (0)i$

① $\tan\alpha = 0 \Rightarrow \alpha = 0$
② 1st quad. $\Rightarrow \theta = 0$

So,

$1 = 1(\cos 0 + i\sin 0)$

Hence, we can write, if

$z^3 = 1$

$z = (1)^{1/3}$

$z = (\cos 0 + i\sin 0)^{\frac{1}{3}}$

Using DMT,

$p=1 \quad q=3 \quad \theta = 0$

$$z = \cos\left(\frac{p\theta+2k\pi}{q}\right) + i\sin\left(\frac{p\theta+2k\pi}{q}\right)$$

$$\Rightarrow \boxed{z = \cos\left(\frac{2k\pi}{q}\right) + i\sin\left(\frac{2k\pi}{q}\right)} \quad (\because \theta = 0)$$

$(k = 0, 1, 2)$

$k=0 \Rightarrow z_1 = \cos 0 + i\sin 0 = e^{i(0)} = 1$

$k=1 \Rightarrow z_2 = \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) = e^{i\left(\frac{2\pi}{3}\right)} = -\frac{1}{2} + \frac{\sqrt{3}i}{2}$ (ω)

$k=2 \Rightarrow z_3 = \cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right) = e^{i\left(\frac{4\pi}{3}\right)} = -\frac{1}{2} - \frac{\sqrt{3}i}{2}$ (ω^2)

Summary :⇒

A cubic equation has three roots.

$z^3 - 1 = 0$ (1) $\omega = -\frac{1}{2} + \frac{\sqrt{3}i}{2}$; $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}i}{2}$

Shortcut to find roots

$z^3 - 1 = (z-1)(z^2+z+1) = 0$

$z-1=0$ | $z^2+z+1=0$

$z=1$ | $z = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$

NOTE:

1) Eqn whose roots are ω & ω^2: $x^2 + x + 1 = 0$ — ω, ω^2

2) Eqn whose roots are $-\omega$ & $-\omega^2$: $x^2 - x + 1 = 0$ — $-\omega$, $-\omega^2$

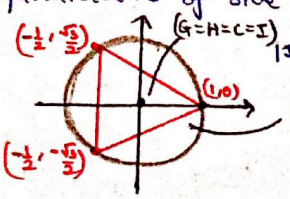
Properties of cube root of unity :⇒

1) $\boxed{\omega^3 = 1}$ ⇒ If power of ω is multiple of 3, then it is equal to 1.

2) $\boxed{1 + \omega + \omega^2 = 0}$

3) $1 + \omega^r + (\omega^2)^r = \begin{cases} 3\,; & r = \text{multiple of } 3 \\ 0\,; & r \neq \text{multiple of } 3 \end{cases}$

4) All three roots lie on a circle of radius 1 unit, on Argand plane. And, form equilateral $\triangle$ of side $\sqrt{3}$.



$|1| = |\omega| = |\omega^2| = 1$

side $= \sqrt{3}$

Area $= \frac{3\sqrt{3}}{4}$

5) All the centres (centroid, circumcentre, orthocentre, Incentre) of equilateral $\triangle$ lies at the origin $(0,0)$

Ques Find value of $\left(\frac{-1+\sqrt{3}i}{1-i}\right)^{30}$.

Sol^n

$$z = \frac{\left(\frac{-1+\sqrt{3}i}{2}\right)^{30}\cdot 2^{30}}{\left(1-i\right)^{30}} = \frac{\omega^{30}\cdot 2^{30}}{(-2i)^{15}}$$

$$z = -\frac{1\cdot 2^{15}\cdot 2^{15}}{2^{15}(-i)} \times \frac{i}{i} \Rightarrow -2^{15}i \quad \text{Ans}$$

Ques If $K = \frac{(-1+\sqrt{3}i)^{21}}{(1-i)^{24}} + \frac{(1+\sqrt{3}i)^{21}}{(1+i)^{24}}$ and $n = [|K|]$ where $[\cdot] \to$ Greatest integer function, then find value of $\sum_{j=0}^{n+5}(j+5)^2 - \sum_{j=0}^{n+5}(j+5)$.

Sol^n

$$K = \frac{\left(\frac{-1+\sqrt{3}i}{2}\right)^{21}\cdot 2^{21}}{(1-i)^{24}} + \frac{\left(\frac{1+\sqrt{3}i}{2}\right)^{21}\cdot 2^{21}}{(1+i)^{24}}$$

$$K = \frac{\omega^{21}\cdot 2^{21}}{(-2i)^{12}} + \frac{(-\omega^2)^{21}\cdot 2^{21}}{(2i)^{12}}$$

$$K = \frac{2^{21}}{2^{12}\cdot i^{12}} + \frac{(-1)(2^{21})}{2^{12}\cdot i^{12}} \Rightarrow 0$$

So

$n = [|0|] = 0$

Hence,

$S = (5^2+6^2+7^2\dots+10^2) - (5+6+7\dots+10)$

$= 330$ Ans

Ques If a and b are real numbers such that $(2+\alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1+\sqrt{3}i}{2}$, then find $a+b$.

Sol^n $(2+\alpha)^4 = a+b\alpha$ | $\alpha = -\frac{1+\sqrt{3}i}{2}$

$(2+\omega)^4 = a+b\omega$ | $\Rightarrow \alpha = \omega$

L.H.S : $(2+\omega)^4 = \left((2+\omega)^2\right)^2 \Rightarrow (4+4\omega+\omega^2)^2$

$= (4+4\omega-1-\omega)^2$ $\left[\because 1+\omega+\omega^2 = 1,\ \omega^2 = 1-\omega\right]$

$= (3+3\omega)^2$

$= 9 + 18\omega + 9\omega^2 \Rightarrow 9 + 18\omega + 9(1-\omega)$

$= 9\omega$

So,

$0 + 9\omega = a + b\omega \Rightarrow \boxed{a+b=9}$ Ans

nth Root of Unity:

Let, $z^n = 1 \Rightarrow z = (1)^{1/n} = (\cos 0 + i\sin 0)^{\frac{1}{n}}$

Using DMT

$p=1 \quad q=n \quad \theta = 0$

$$\Rightarrow \boxed{z = \cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right)}$$

$(k = 0, 1, 2, 3 \dots n-1)$

$k=1 \Rightarrow z_1 = e^{i(0)} = 1$

$k=2 \Rightarrow z_2 = e^{i\left(\frac{2\pi}{n}\right)} = \alpha$

$K=3 \Rightarrow z_3 = e^{i\left(\frac{2\pi}{n}\right)} = \alpha^2$

$K=4 \Rightarrow z_4 = e^{i\left(\frac{3\pi}{n}\right)} = \alpha^3$

$\vdots \qquad \vdots$

$K=n-1 \Rightarrow z_n = e^{i\left(\frac{2(n-1)\pi}{n}\right)} = \alpha^{n-1}$

Summary :→

n^{th} equation of nth degree have n-roots that can be given by

$z^n - 1 = 0 \Rightarrow 1, \alpha, \alpha^2, \ldots, \alpha^{n-1}$ where $\alpha = e^{i\left(\frac{2\pi}{n}\right)}$

* (1) Real Root = 1
* (n-1) complex Roots = $\alpha, \alpha^2, \alpha^3 \ldots \alpha^{n-1}$
* Total no. of Roots = n
* All the roots are in GP with $r = \alpha$

Properties of nth roots of Unity :→

1) **Sum of Roots**

$$1 + \alpha + \alpha^2 + \ldots + \alpha^{n-1} = 0$$

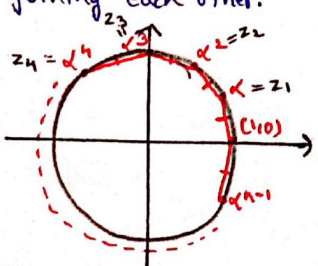
2) **Product of Roots**

$$1 \times \alpha \times \alpha^2 \ldots \times \alpha^{n-1} = \begin{cases} 1 ; & n = \text{odd} \\ -1 ; & n = \text{even} \end{cases}$$

3) **Sum of each root raised to power (r)**

$$1^r + (\alpha)^r + (\alpha^2)^r + \ldots + (\alpha^{n-1})^r = \begin{cases} n ; & r = \text{multiple of } n \\ 0 ; & r \neq \text{multiple of } n \end{cases}$$

4) All the roots lie on a circle of radius = 1, and form a n-sided regular polygon on joining each other.



All sides are equal

i.e. $|z_2 - z_1| = |z_3 - z_2| = |z_4 - z_3| = \ldots$

Ques Find $\sum_{K=1}^{12} \left[\sin\left(\frac{2K\pi}{13}\right) - i\cos\left(\frac{2K\pi}{13}\right)\right]$

Soln

$$S = \frac{12}{\sum_{K=1}}\ -i\left[\cos\left(\frac{2K\pi}{13}\right) + i\sin\left(\frac{2K\pi}{13}\right)\right]$$

$S = -i\left(1 + \alpha + \alpha^2 + \alpha^3 \ldots + \alpha^{12}\right) \Rightarrow -i(-1)$

$\boxed{S = -i}$ Ans

Ques For any integer K, let $a_K = \cos\left(\frac{K\pi}{4}\right) + i\sin\left(\frac{K\pi}{4}\right)$. Then find the value of expression

$$\sum_{K=1}^{12} |a_{K+1} - a_K| \Big/ \sum_{K=1}^{3} |a_{4K-1} - a_{4K-2}|$$

Soln $a_K = \cos\left(\frac{2K\pi}{14}\right) + i\sin\left(\frac{2K\pi}{14}\right) = e^{i\left(\frac{2K\pi}{14}\right)}$

Now,

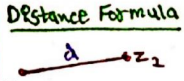
$$S = \frac{|a_{13} - a_{12}| + |a_{12} - a_{11}| \ldots + |a_2 - a_1|}{|a_{11} - a_{10}| + |a_7 - a_6| + |a_3 - a_2|}$$

Since, diff of consecutive roots (mod of) give us side of polygon formed by joining them.

$$S = \frac{12a}{3a} \Rightarrow 4 \text{ Ans}$$

Geometry of Complex Numbers :

Distance Formula :→



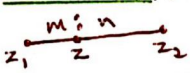
$\boxed{d = |z_2 - z_1|}$

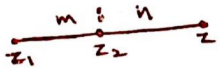
So, $|z+1| \Rightarrow$ Distance b/w z & -1

$|z-2+3i| \Rightarrow$ Distance b/w z & (2-3i)

$|z-i| \Rightarrow$ Distance b/w z & i

Section Formula :→

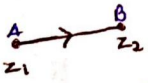




$\boxed{z = \frac{mz_2 + nz_1}{m+n}}$ (Internal division)

$\boxed{z = \frac{mz_2 + nz_1}{m-n}}$ (External Division)

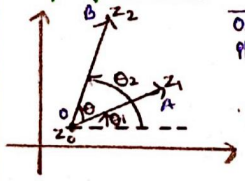
Complex Numbers as Vectors :→



$\boxed{\overrightarrow{AB} = z_2 - z_1}$

$|\overrightarrow{AB}| = |z_2 - z_1|$ → Same as distance formula.

Concept of Rotation :→



$\overrightarrow{OA}$ rotated by θ, also its magnitude is increased

$z_1 - z_0 = |z_1 - z_0| e^{i\theta_1}$

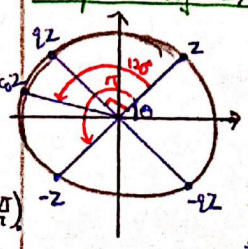
$z_2 - z_0 = |z_2 - z_0| e^{i\theta_2}$

$$\frac{z_2 - z_0}{z_1 - z_0} = \frac{|z_2 - z_0|}{|z_1 - z_0|} e^{i(\theta_2 - \theta_1)}$$

$$\Rightarrow \boxed{\frac{z_2 - z_0}{z_1 - z_0} = \frac{|z_2 - z_0|}{|z_1 - z_0|} e^{i\theta}}$$

Final / Initial $= \frac{|\text{Final}|}{|\text{Initial}|} e^{i\theta}$; θ is ACW $\Rightarrow$ +ve ; θ is CW $\Rightarrow$ -ve

Multiplication of -1, i, -i, ω & -ω :→



Let, $z = re^{i\theta}$

* $-z = 1 \cdot e^{i(\theta + \pi)}$
* $iz = 1 \cdot e^{i(\theta + \pi/2)}$
* $-iz = 1 \cdot e^{i(\theta - \pi/2)}$
* $\omega z = 1 \cdot e^{i(\theta + 2\pi/3)}$
* $-\omega z = 1 \cdot e^{i(\theta - 2\pi/3)}$

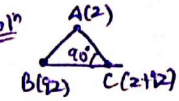
$-1 \Rightarrow$ ~~Decrease~~ Increase by $\pi/180°$

$i, -i \Rightarrow$ Use as use by $\pi/90°$

$\omega, -\omega \Rightarrow$ Use as use by $2\pi/120°$

Ques The area of the Δ with vertices A(z), B(iz) and C(z+iz) is _____.

Soln



Since Δ is a right angled Δ,

$\text{Area} = \frac{1}{2} \times AC \times BC$

$= \frac{1}{2} \times |iz| \times |z|$

$\boxed{\text{Area} = \frac{1}{2}|z|^2}$ Ans $\quad (\because |i| = 1)$

Important Results :→

If z_1, z_2, z_3 are the vertices of an 'Equilateral Triangle' then,

$$\boxed{z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_1z_3}$$

And, if z_0 is the circumcentre of the triangle then

$$\boxed{3z_0^2 = z_1^2 + z_2^2 + z_3^2}$$

Ques Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1, z_2 form an equilateral Δ with origin. Then value of $|a|$ is _____.

Soln Vertices of eq $\Delta \equiv z_1, z_2$ & (0,0)

So,

$z_1^2 + z_2^2 + (0)^2 = z_1z_2 + 0 + 0$

$(z_1 + z_2)^2 - 2z_1z_2 = z_1z_2$

$(-a)^2 = 3(12)$

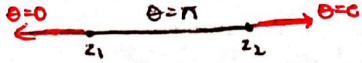
$\Rightarrow \boxed{|a| = 6}$ Ans

Standard Locus Involving Argument :

$$\boxed{\arg\left(\frac{z - z_1}{z - z_2}\right) = \theta}$$

→ It is angle b/w two CN $(z - z_1)$ & $(z - z_2)$.

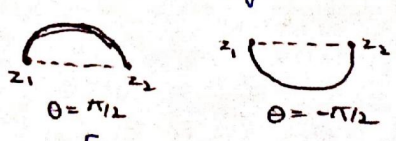
(i) **If $\theta = 0$ or π :→** Locus is a straight line.



$\theta = \pi \Rightarrow$ Line segment joining z_1 & z_2

$\theta = 0 \Rightarrow$ Rays in opp. dir^n, originating from z_1 & z_2

(ii) **If $\theta = \pm\pi/2$:→** Locus is semi-circle, kyonki semi-circle mein hi angle 90° hota hai.



[$z_1, z_2 \Rightarrow$ Diametric ends]

Complex Number-3

(iii) When $\theta \neq 0, \pi, \pm\pi/2$:→

If θ has some acute or obtuse value, besides the above two cases; then locus is major arc or minor ~~arc~~ arc respectively.



$\theta \to$ Acute $< 90°$



$\theta \to$ Obtuse $> 90°$

Ques If $z_1 = 2+3i$, $z_2 = 3-2i$ and $z_3 = -1-2\sqrt{3}i$, then which of the following is true?

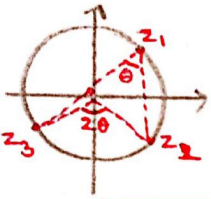
(A) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3-z_1}{z_2-z_1}\right)$ (B) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_2}{z_1}\right)$

(C) $\arg\left(\frac{z_3}{z_2}\right) = 2\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)$ (D) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3}{z_1}\right)$

Sol^n If we calculate modulus of CNs then,

$$|z_1| = |z_2| = |z_3| = \sqrt{13}$$

i.e, all complex numbers lie on a circle of radius $\sqrt{13}$ units.



$\arg\left(\frac{z_3}{z_2}\right) \Rightarrow$ Angle b/w z_3 & z_2 $= 2\theta$

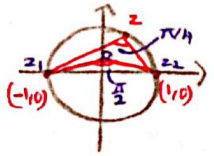
$\arg\left(\frac{z_3-z_1}{z_2-z_1}\right) \Rightarrow$ Angle b/w these two vectors $= \theta$

So, $\boxed{\arg\left(\frac{z_3}{z_2}\right) = 2\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)}$ (C) Ans

Ques The eqn $\arg\left(\frac{z-1}{z+1}\right) = \pi/4$ represents a circle with ___ radius & ___ centre.

Sol^n $\arg\left(\frac{z-1}{z-(-1)}\right) = \pi/4$





$\tan 45° = \frac{MZ_2}{OM} = 1 \Rightarrow \boxed{OM = 1}$ i.e $\boxed{\text{Centre} \equiv (0,1)}$

Ans $\boxed{\text{Radius} \equiv \sqrt{2}}$

Equation of Straight Line :

To find analogy with our cartesian form, just replace

$y \longrightarrow z$

$x \longrightarrow \bar{z}$

$$\boxed{\bar{a}z + a\bar{z} + b = 0}$$ where, $b \in R$, $a \to$ Some CN

Complex Slope :

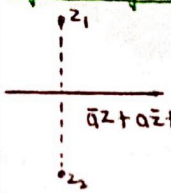
$$\omega = \frac{-a}{\bar{a}} = \frac{-\text{coeff of } \bar{z}(z)}{\text{coeff of } z(y)}$$

* If ω_1 & ω_2 are complex slopes of two straight lines, then

Parallel Lines $\Rightarrow \boxed{\omega_1 = \omega_2}$

Perpendicular Lines $\Rightarrow \boxed{\omega_1 + \omega_2 = 0}$

Reflection of a Point in Straight Line :→



① Write eqn of st. line as $\bar{a}z + a\bar{z} + b = 0$

② Substitute,

First $z \to z_2$

Second $\bar{z} \to \bar{z}_1$

i.e $\boxed{\bar{a}z_2 + a\bar{z}_1 + b = 0}$

We know everything except z_2 in this eqn, hence we will get z_2 by it.

Eg: Find image of the point $P(2,-1)$ in the line $(1+i)z - (1-i)\bar{z} + 2i = 0$.

Sol^n Eqn of line is not in standard form b/c 'b' is not Real here.

$$(1+i)z - (1-i)\bar{z} + 2i = 0$$

i.e $\quad (i+i^2)z - (i-i^2)\bar{z} + 2i^2 = 0$

$$(i-1)z_2 - (i+1)\bar{z}_1 - 2 = 0$$

$$\Rightarrow (i-1)z_2 - (i+1)(1+i) - 2 = 0$$

$$z_2 = \frac{(1+i)^2+2}{(i-1)} \Rightarrow \frac{i^2+2i+1+2}{(i-1)}$$

$$z_2 = \frac{2(i+1)}{(i-1)} \times \frac{(i+1)}{(i-1)} \Rightarrow -2i$$

So, $\boxed{z_2 \equiv (0,-2)}$ Ans

Distance of point from Line :→



$$\boxed{d = \frac{|\bar{a}z_1 + a\bar{z}_1 + b|}{2|a|}}$$

↳ Substitute z_1 i.e eqn of line & divide by twice of mod of 'a' or '$\bar{a}$'

Ques Let z be a CN such that $z^2 = i|z|^2$ then z lies on ___.

Sol^n $z.z = i.z.\bar{z} \Rightarrow (x+iy) = i(x-iy)$

z lies on line $x = y$ ← [$x+iy = y+ix$]

Different forms of Eqn of Circle :

1) $\boxed{|z| = r}$ centre $\equiv (0,0)$, radius $= r$

2) $\boxed{|z-z_0| = r}$ centre $\equiv z_0$, radius $= r$

3) $\boxed{z\bar{z} + \bar{a}z + a\bar{z} + b = 0}$ (Line ki eqn mein $z\bar{z}$ add karo) centre $\equiv (-a)$, $\boxed{\text{radius} = \sqrt{|a|^2 - b}}$ ($a \in C$ & $b \in R$)

Eg: Find centre and radius of

$$z\bar{z} - (3-4i)z - (3+4i)\bar{z} + 9 = 0$$

Sol^n $z\bar{z} + (-3+4i)z + (-3-4i)\bar{z} + 9 = 0$

$$z\bar{z} + \bar{a}z + a\bar{z} + b = 0$$

Centre $\equiv (3,4)$ | $r = \left(\sqrt{3^2+4^2} - 9\right)^{1/2}$

$r = \sqrt{16} = 4$ Ans

Ques If eqn $a|z|^2 + \bar{\alpha}z + \alpha\bar{z} + d = 0$ represent a circle where 'a' & 'd' are real constants. Then find the necessary cond^n for circle to exist.

Sol^n $a|z|^2 + \bar{\alpha}z + \alpha\bar{z} + d = 0$

$$z.\bar{z} + \frac{\bar{\alpha}}{a}.z + \frac{\alpha}{a}.\bar{z} + \frac{d}{a} = 0$$

For circle to exist, radius must exist

i.e $r = \sqrt{\left|\frac{\alpha}{a}\right|^2 - \frac{d}{a}} \Rightarrow \sqrt{\frac{|\alpha|^2 - ad}{a^2}}$

$$\Rightarrow \frac{|\alpha|^2 - ad}{a^2} > 0 \Rightarrow \boxed{|\alpha|^2 - ad > 0} \text{ —①}$$

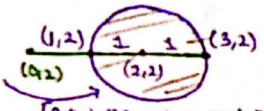
Also, $\boxed{a \neq 0}$ —② Ans

Ques If for the CN z satisfying $|z-2-2i| \leq 1$, the maximum value of $|3iz+6|$ is attained at $(a+ib)$, then $a+b$ is ___.

Sol^n It is given

$|z-z_0| \leq 1$

$C \equiv (2,2)$ & $r = 1$



[Point lie ON or IN the circle]

Now, $|z'| = |3iz+6| \Rightarrow \left|3i\left(z+\frac{6}{3i}\right)\right|$

$= |3i| \cdot |z-2i| \Rightarrow 3|z-2i|$ max.

We Know,

$|z-2i| \Rightarrow$ distance b/w z & $2i$ $(0,2)$

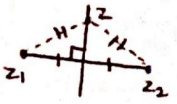
(Distance b/w $(0,2)$ & z can be max. only if $(z = 3+2i)$)

Thus, $\boxed{a+b = 5}$ Ans

Standard Locus Involving Modulus :

* Type-01 :→

$$\boxed{\left|\frac{z-z_1}{z-z_2}\right| = k}$$



bisector

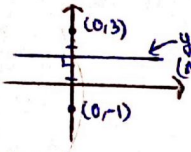
↳ If $k = 1 \Rightarrow$ Locus is bar line ~~segment~~ of line joining z_1 & z_2

↳ If $k \neq 1 \Rightarrow$ Locus is a circle of form $(z\bar{z} + \bar{a}z + a.\bar{z} + b = 0)$

Ques Let z & w be two CN such that $\left|\frac{z+i}{z-3i}\right| = 1$, $w = z\bar{z} - 2z + 2$ and $Re(w)$ has min. value.

then, min. value of $n \in N$ for which w^n is real, is equal to ____.

Sol^n $\left|\frac{z-(-i)}{z-(3i)}\right| = 1$ → Locus bisector of line joining $(0,-1)$ & $(0,3)$



So, $\boxed{z = x + i(1)}$

Now, $w = z\bar{z} - 2z + 2$

$w = (x+i)(x-i) - 2(x+i) + 2$

$w = x^2 - xi + ix - i^2 - 2x - 2i + 2$

$w = (x^2 - 2x + 3) + (-2)i$

Since, $Re(w) = Minimum \Rightarrow \frac{d(x^2-2x+3)}{dx} = 0$

$\Rightarrow 2x - 2 = 0$

$\boxed{x = 1}$

$Re(w)_{min} = 1 - 2 + 3 = 2$

Hence,

$w = (2 - 2i) \Rightarrow 2(1-i)$

$w^n = 2^n (1-i)^n \Rightarrow 2^n [(1-i)^2]^{n-2}$

$w^n = 2^n (-2i)^{n-2} \Rightarrow 2^n ((-2i)^2)^{n-4}$

$w^n = 2^n (-4 \times -1)^{n-4} \Rightarrow 2^n (4)^{n-4}$

Thus, $\boxed{n_{min} = 4}$ Ans

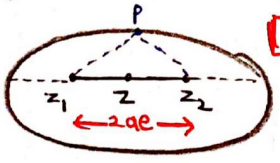
***Type-02 :→**

$\boxed{|z-z_1| + |z-z_2| = K}$

→ If $K < |z_1 - z_2| \Rightarrow$ No real locus

→ If $K = |z_1 - z_2| \Rightarrow$ Line joining z_1 & z_2

→ If $K > |z_1 - z_2| \Rightarrow$ Ellipse.



$\boxed{PS_1 + PS_2 = 2a}$

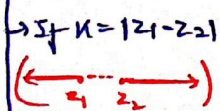
*Whether the ellipse will be horizontal or vertical, it depends upon coordinates of z_1 & z_2.

***Type-03 :→**

$\boxed{|z-z_1| - |z-z_2| = K}$

→ If $K > |z_1 - z_2| \Rightarrow$ No Real locus

→ If $K = |z_1 - z_2| \Rightarrow$ Locus is Rays emerging out of z_1 & z_2 in opp dir^n

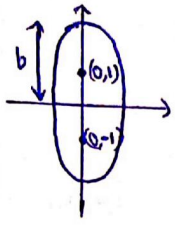


→ If $K < |z_1 - z_2| \Rightarrow$ Hyperbola $\boxed{|PS_1 - PS_2| = 2a}$

Que If z is a complex number satisfying the equation $|z+i| + |z-i| = 8$, on the complex plane, then maximum value of $|z|$ is ____.

Sol^n $|z+i| + |z-i| = 8$

↳ Represents an ellipse; because 8 is greater than $|i - (-i)| \Rightarrow 2$



Now, $|z|$ will be maximum only if z lies on any of the vertices.

i.e. $|z| = b$

$\Rightarrow \boxed{|z|_{max} = 4}$ Ans

Log of Complex Number :

Step-01 :→ Write given CN in Euler's form

i.e. $z = re^{i\theta}$

Step-02 :→ Take natural log of both sides and separate r & $e^{i\theta}$.

$\ln z = \ln(re^{i\theta}) \Rightarrow \ln r + \ln e^{i\theta}$

$\Rightarrow \boxed{\ln z = \ln r + i(\theta + 2k\pi)}$

Eg : $\ln(1+i)$

Sol^n $\ln z = \ln(1+i)$

$= \ln(\sqrt{2} e^{i(\pi/4)})$

$= \ln\sqrt{2} + \ln e^{i\theta(\pi/4)}$

$= \frac{1}{2}\ln 2 + i\left(\frac{\pi}{4}\right)$ Ans $+2k\pi$

Eg : $\ln(-5)$

Sol^n $z = -5 + (0)i \rightarrow \boxed{\theta = \pi}$

$\ln(z) = \ln(5e^{i\pi})$

$= \ln 5 + \ln e^{i\pi}$

$= \ln 5 + i(\pi + 2k\pi)$ Ans